



**PORT HACKING HIGH SCHOOL**

Student name:.....

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Centre Number

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Student Number

**2014**  
**Higher School Certificate**  
**Trial Examination**

# Mathematics

## General Instructions:

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black pen or blue pen
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- In Questions 11 – 16, show relevant mathematical reasoning and/or calculations
- Clearly identify the question you are answering on each writing booklet

**Total marks – 100**

## Section I

10 Marks

- Attempt Questions 1 - 10
- Answer on the multiple choice answer sheet provided
- Allow about 15 minutes for this section

## Section II

90 Marks

- Attempt Questions 11 – 16
- Allow about 2 hours and 45 minutes for this section

|                 |             |
|-----------------|-------------|
| Multiple Choice | /10         |
| Question 11     | /15         |
| Question 12     | /15         |
| Question 13     | /15         |
| Question 14     | /15         |
| Question 15     | /15         |
| Question 16     | /15         |
| <b>Total</b>    | <b>/100</b> |

## Section I

10 marks

Attempt Questions 1 - 10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1-10

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- 1  $4^2 \times 5^3 \times (5 \times 7^2)^4 \div 4^{-5}$  is equivalent to:
- (A)  $4^7 \times 5^7 \times 7^8$   
(B)  $4^{-3} \times 5^7 \times 7^6$   
(C)  $4^7 \times 5^7 \times 7^6$   
(D)  $4^{-3} \times 5^7 \times 7^8$
- 2 For the function  $f(x) = \log_e x$ , the value of  $\lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$  is:
- (A)  $e^x$   
(B)  $\log_e x$   
(C)  $\frac{1}{x}$   
(D)  $e \log_e x$
- 3 What is the size of each interior angle in a regular 20-sided polygon?
- (A)  $20^\circ$   
(B)  $162^\circ$   
(C)  $180^\circ$   
(D)  $3240^\circ$
- 4 Which of the following describes the locus of all points which are equidistant from  $A(2, 3)$  and  $B(4, 5)$
- (A)  $(x - 3)^2 + (y - 4)^2 = 1$   
(B)  $x + y - 7 = 0$   
(C)  $(x - 3)^2 = 4(y - 4)$   
(D)  $x - y + 1 = 0$

5 What is the gradient of any line perpendicular to  $x + 3y = 2$ ?

- (A)  $-3$
- (B)  $-\frac{1}{3}$
- (C)  $\frac{1}{3}$
- (D)  $3$

6 For the function  $y = \left| \frac{2}{x-3} \right|$ , the Range is:

- (A)  $y \neq 2$
- (B)  $y \neq 0$
- (C)  $y > 0$
- (D)  $x \neq 3$

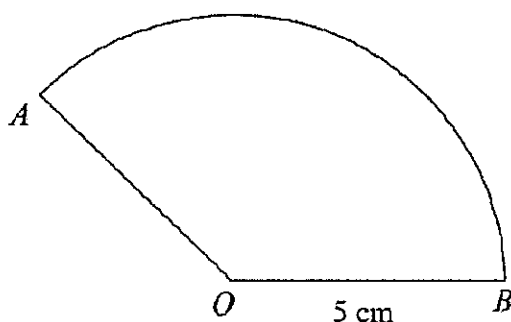
7 Which is not equal to  $\cos 40^\circ$ ?

- (A)  $\cos 400^\circ$
- (B)  $\cos(-40^\circ)$
- (C)  $\cos 220^\circ$
- (D)  $\cos 320^\circ$

8 If  $f(x) = 4(7 - 2x)^3$  then  $f'(x) =$

- (A)  $3(7 - 2x)^2$
- (B)  $-12(7 - 2x)^2$
- (C)  $21(7 - 2x)^2$
- (D)  $-24(7 - 2x)^2$

9  $AOB$  is a sector of a circle, centre  $O$  and radius 5 cm. The sector has an area of  $10\pi \text{ cm}^2$ .



Not to scale

What is the arc length of the sector?

- (A)  $2\pi \text{ cm}$
- (B)  $4\pi \text{ cm}$
- (C)  $6\pi \text{ cm}$
- (D)  $10\pi \text{ cm}$

10 For the sequence  $1, \frac{1}{x}, \frac{1}{x^2}, \dots$ , the  $n$ th term is:

(A)  $x^{1-n}$

(B)  $\frac{1}{x^n}$

(C)  $\frac{n}{x}$

(D)  $\frac{1}{nx}$

**END OF SECTION I**

## Section II

90 marks

Attempt Questions 11 – 16

Allow about 2 hours and 45 minutes for this section

Answer each question in the appropriate writing booklet.

All necessary working should be shown in every question.

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### Question 11 (15 marks)

Marks

- (a) Evaluate  $\tan 0.84^c$  correct to 3 significant figures. 1
- (b) If  $a + \sqrt{b} = 4(3 + 2\sqrt{5})$ , find the value of  $a$  and  $b$ . 2
- (c) Find the exact value(s) of  $k$  for which  $x^2 + 2kx + k + 2 = 0$  has real roots. 2
- (d) The roots of the equation  $2x^2 - x - 15 = 0$  are  $\alpha$  and  $\beta$ . Find the value of:
- (i)  $\alpha + \beta$  1
- (ii)  $\alpha\beta$  1
- (iii)  $\frac{1}{\alpha^2} + \frac{1}{\beta^2}$  2
- (e) Differentiate  $2x^3 \tan 4x$  2
- (f) Evaluate  $\int_0^{\frac{\pi}{3}} \sin 3x \, dx$  2
- (g) Solve the equation  $2 \cos x + 1 = 0$  in the domain  $0 \leq x \leq 2\pi$ . 2

END OF QUESTION 11

**Question 12 (15marks)****START A NEW BOOKLET****Marks**

(a) Solve the following equations:

(i)  $|5x - 2| = |2x + 3|$  2

(ii)  $2^{x+1} = 8^x$  1

(b) For the parabola with equation  $16y = x^2 - 4x - 12$ , find the:

(i) The coordinates of the vertex. 2

(ii) The coordinates of the focus. 1

(iii) The equation of the directrix. 1

(iv) The equation of the tangent to the parabola at the point  $\left(0, \frac{-3}{4}\right)$  2(c) If the gradient of a function is given by  $f'(x) = \frac{2x}{x^2-3}$ , find  $f(x)$  given that  $f(2) = 4$  2

(d) The fifth term of an arithmetic series is 42 and the twelfth term is 21.

(i) Find the common difference. 2

(ii) Find the sum of the first 80 terms. 1

(iii) Find the smallest value of  $n$ , such that  $S_n < 0$  1**END OF QUESTION 12**

- (a) Alex ( $A$ ) and Bella ( $B$ ) leave from point  $O$  at the same time. Alex travels at 20 km/h along a straight road in the direction  $085^\circ$  T. Bella travels at 25 km/h along another straight road in the direction  $340^\circ$  T.

Draw a diagram to represent this information.

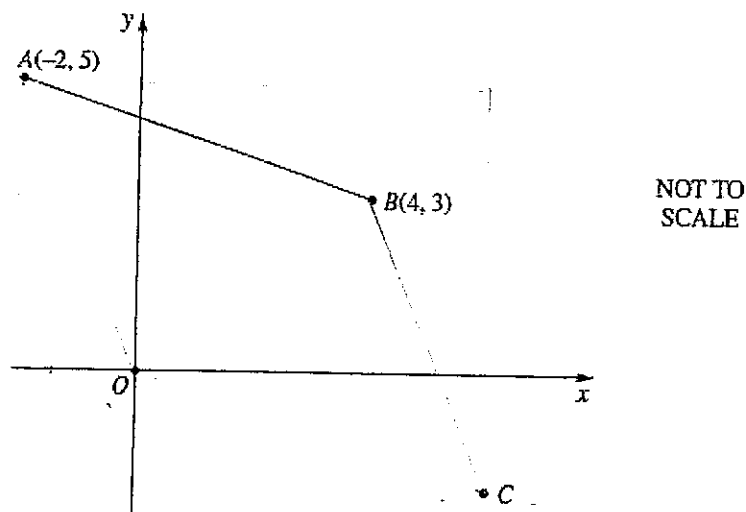
- (i) Show that  $\angle AOB$  is  $105^\circ$  where  $\angle AOB$  is the angle between the directions taken by Alex and Bella.

1

- (ii) Find the distance Alex and Bella are apart to the nearest kilometre after two hours.

2

(b)



The diagram shows the points  $A(-2, 5)$ ,  $B(4, 3)$  and  $O(0, 0)$ . The point  $C$  is the fourth vertex of the parallelogram  $OACB$ .

- (i) Show that the equation of  $AB$  is  $x + 3y - 13 = 0$
- (ii) Find the exact perpendicular distance from  $O$  to the line  $AB$ .
- (iii) Find the coordinates of  $C$ , such that  $OACB$  is a parallelogram.
- (iv) Calculate the exact length of  $OC$ .
- (v) Calculate the area of parallelogram  $OACB$ .

2

1

1

1

1

(c) Let  $f(x) = x^3 - 3x^2 - 9x + 22$

- |       |                                                                                           |   |
|-------|-------------------------------------------------------------------------------------------|---|
| (i)   | Find the coordinates of the stationary points and determine their nature.                 | 2 |
| (ii)  | Find the coordinates of the point of inflexion.                                           | 1 |
| (iii) | Sketch the graph of $y = f(x)$ for $-3 \leq x \leq 6$ , showing all relevant information, | 2 |
| (iv)  | Find the maximum value of $f(x)$ , for $-3 \leq x \leq 6$                                 | 1 |

**END OF QUESTION 13**

**Question 14 (15marks)**

**START A NEW BOOKLET**

**Marks**

- (a) Evaluate:  $\lim_{x \rightarrow \infty} \frac{x^4 + 3x^2 + 2}{5x^4 + 1}$  1
- (b) Evaluate the expression  $\log_4 22$  to 3 decimal places. 1
- (c) Find  $\frac{dy}{dx}$  if:
- (i)  $y = \frac{e^x}{x}$  2
- (ii)  $y = x \log_e 2x$  2
- (d) Evaluate  $\sum_{n=1}^{\infty} 3^{1-n}$  2
- (e) Simplify the expression  $\frac{1}{2} \log_e(x+3) - \log_e \frac{x}{2}$  2
- (f) (i) Sketch the function  $y = \log_e x$  1
- (ii) Write down the domain of this function. 1
- (iii) By using calculus, show that the function is always increasing. 1
- (iv) Shade the region on your graph that satisfies: 2
- $$y \geq \log_e x \quad \text{and} \quad (x-1)^2 + y^2 < 1$$

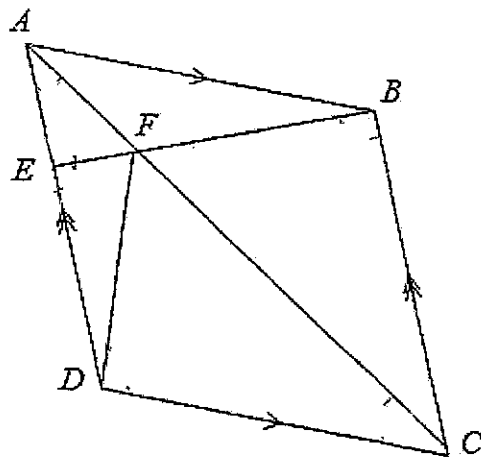
**END OF QUESTION 14**

- (a) *Temp4U* is an employment agency which specialises in contracting temporary employees. They have analysed the number of job applications received over the last five years. They found that the demand ( $D$ ), measured in hundreds, for temporary employment at time ( $t$  years) is given by the function:

$$D(t) = 4 \sin\left(\frac{\pi}{4}t\right) + 7$$

- (i) State the amplitude and period of  $D(t)$  and sketch its graph for the first twelve years. 3
- (ii) Using your graph, approximate the times in the first 12 years when the demand will be at its peak. 1

(b)



ABCD is a rhombus,  $BE$  is perpendicular to  $AD$  and intersects  $AC$  at  $F$ .

Copy the diagram into your workbook.

- (i) Explain why  $\angle BCA = \angle DCA$ . 1
- (ii) Prove that  $\triangle BFC$  and  $\triangle DFC$  are congruent. 2
- (iii) Show that  $\angle FBC$  is a right angle. 1
- (iv) Hence or otherwise find the size of  $\angle FDC$ , giving reasons. 1

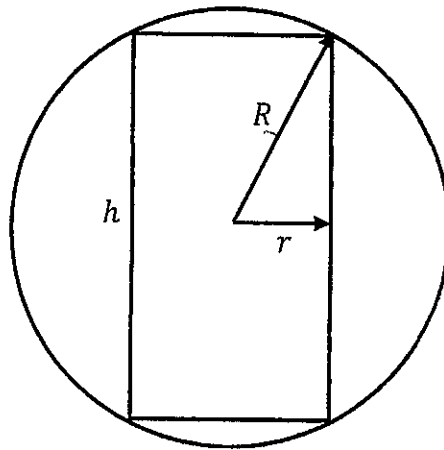
Question 15 continues on the next page

- (c) Given that  $\int_0^6 (x + k) \, dx = 30$ , and  $k$  is a constant, find the value of  $k$ .

2

- (d) A metal worker is required to cut a cylinder from a solid sphere of metal of radius  $R = 4$  cm. The cylinder has a radius  $r$  cm and a height  $h$  cm.

The diagram below shows a cross section of the sphere and cylinder.



NOT TO  
SCALE

- (i) Show that the volume  $V \text{ cm}^3$ , of the cylinder is given by

$$V = \frac{1}{4}\pi h(64 - h^2)$$

2

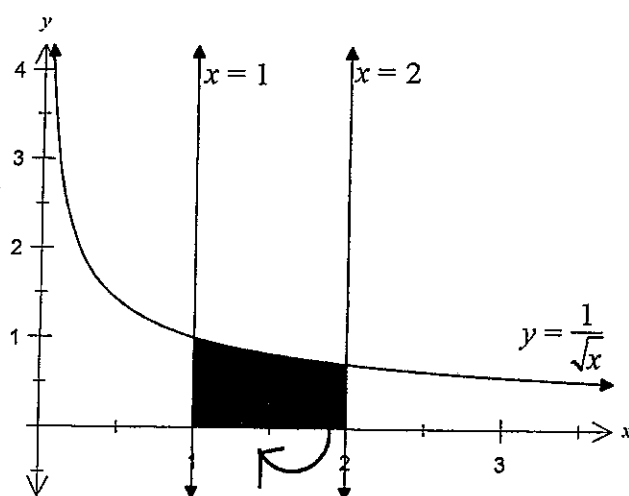
- (ii) Hence, find the exact value of  $h$ , that gives the maximum volume of the cylinder.

2

**END OF QUESTION 15**

**Question 16 (15marks)****START A NEW BOOKLET****Marks**

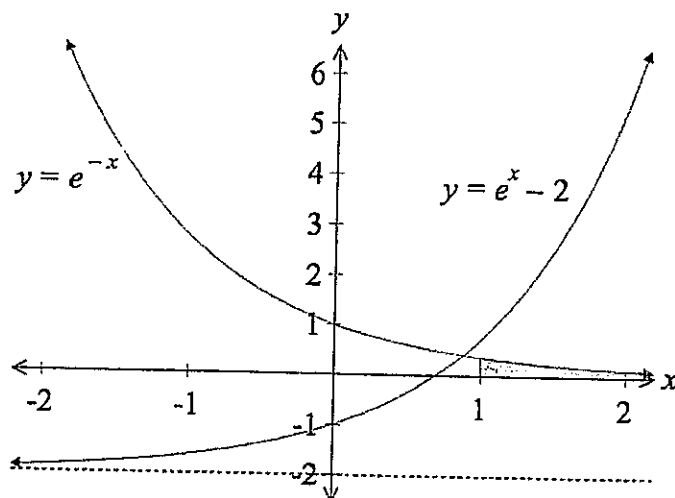
- (a) Evaluate  $\int_0^4 \sqrt{2+x^4} dx$  using Simpson's rule with 3 function values.  
Answer to 2 decimal places. 2
- (b) An arithmetic sequence is defined by  $S_n = 3n^2 - 11n$ .
- (i) Write down the expression for  $S_{n-1}$  1
- (ii) Hence or otherwise write down the expression for the  $n$ th term,  $T_n$  1
- (c) (i) If  $y = \log_e(\cos x)$ , find  $\frac{dy}{dx}$  1
- (ii) Hence, find  $\int \tan x dx$ . 1
- (d) The region between the functions  $y = \frac{1}{\sqrt{x}}$ ,  $x = 1$  and  $x = 2$  is rotated about the  $x$ -axis, as shown below. 2



Find the volume of the solid formed.

Question 16 continues on the next page

- (e) The diagram below shows the graphs of  $y = e^x - 2$  and  $y = e^{-x}$



- (i) Find the area between the curves from  $x = 1$  and  $x = 2$ . Leave your answer in terms of  $e$ . 3
- (ii) Show that the curves intersect when  $e^{2x} - 2e^x - 1 = 0$ . 1
- (iii) Find the  $x$  coordinate of the point of intersection of the curves. Answer correct to 3 decimal places. 3

**END OF EXAM**

## STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left( x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left( x + \sqrt{x^2 + a^2} \right)$$

NOTE:  $\ln x = \log_e x$ ,  $x > 0$

1. A 2. C 3. B 4. B 5. D  
6. C 7. C 8. D 9. B 10. A

### Question 11

a) 1.12

b)  $a + \sqrt{b} = 12 + 8\sqrt{5}$   
 $= 12 + \sqrt{320}$

$a = 12, b = 320$

c)  $\Delta = b^2 - 4ac \geq 0$  for real roots  
 $(2k)^2 - 4 \times 1 \times (k+2) \geq 0$

$4k^2 - 4k - 8 \geq 0$

$k^2 - k - 2 \geq 0$

$(k-2)(k+1) \geq 0$

$k \geq 2$  and  $k \leq -1$

d)  $\alpha + \beta = -\frac{b}{a} = \frac{1}{2}$

ii)  $\alpha\beta = \frac{c}{a} = -\frac{15}{2} = -7\frac{1}{2}$

iii)  $\frac{1}{\alpha^2} + \frac{1}{\beta^2} = \frac{\beta^2 + \alpha^2}{\alpha^2 \beta^2}$   
 $= \frac{(\alpha + \beta)^2 - 2\alpha\beta}{(\alpha\beta)^2}$   
 $= \frac{(\frac{1}{2})^2 - 2 \times -7\frac{1}{2}}{(-7\frac{1}{2})^2}$   
 $= \frac{61}{225}$

e)  $y' = v du + u dv$   
 $= 2x^3 \cdot 4 \sec^2 4x + \tan 4x \cdot 6x^2$   
 $= 8x^3 \sec^2 4x + 6x^2 \tan 4x$

f)  $\int_0^{\pi/3} \sin 3x = \left[ -\frac{\cos 3x}{3} \right]_0^{\pi/3}$   
 $= -\frac{1}{3} [-1 - 1] = \frac{2}{3}$

g)  $\cos x = -\frac{1}{2}$  2nd, 3rd quadrant  
 $\cos \pi/3 = \frac{1}{2}$   
 $x = \frac{2\pi}{3}, \frac{4\pi}{3}$

### Question 12

a) i)  $|5x-2| = |2x+3|$   
 $5x-2 = 2x+3$  or  $5x-2 = -(2x+3)$   
 $3x = 5$   $7x = -1$   
 $x = \frac{5}{3}$  or  $x = -\frac{1}{7}$

d) ii)  $2^{x+1} = (2^3)^x$

$x+1 = 3x$

$x = \frac{1}{2}$

b)  $16y = (x-2)^2 - 16$   
 $(x-2)^2 = 16(y+1)$   $a=4$

i) vertex =  $(2, -1)$

ii) focus =  $(2, 3)$

iii) directrix:  $y = -5$

iv)  $y = \frac{x^2}{16} - \frac{x}{4} - \frac{3}{4}$

$y' = \frac{x}{8} - \frac{1}{4} \Rightarrow m = -\frac{1}{4}$  at  $x=0$

$y - y_1 = m(x - x_1)$

$y + \frac{3}{4} = -\frac{1}{4}(x - 0)$

$y = -\frac{x}{4} - \frac{3}{4}$

c)  $f(x) = \log_e(x^2 - 3) + c$

$4 = \log_e(2^2 - 3) + c$

$4 = \log_e 1 + c$

$4 = 0 + c \Rightarrow c = 4$

$f(x) = \log_e(x^2 - 3) + 4$

d) i)  $a + 4d = 42 \dots \dots (1)$

$a + 11d = 21 \dots \dots (2)$

(1) - (2)  $-7d = 21$   $d = -3, a = 54$

ii)  $S_n = \frac{n}{2} [2a + (n-1)d]$   
 $= \frac{80}{2} [2 \times 54 + 79 \times -3]$   
 $= -5160$

iii)  $\frac{n}{2} [2a + (n-1)d] < 0$

$\frac{n}{2} [2 \times 54 + (n-1) \times -3] < 0$

$n [108 - 3n + 3] < 0$

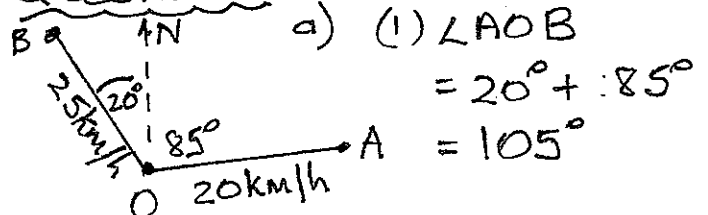
$-3n^2 + 111n < 0$

$3n(-n + 37) < 0$

$n = 0$  (not valid)  $n = 37$

$\therefore n = 38$  if  $S_n < 0$

### Question 13



Q13 cont

a) ii)  $a^2 = b^2 + c^2 - 2bc \cos A$

$$a^2 = 50^2 + 40^2 - 2 \times 50 \times 40 \cos 105^\circ$$

$$= 5063.8$$

$$a = \sqrt{5063.8} = 71 \text{ km}$$

b) i) m of AB =  $\frac{5-3}{-2-4} = -\frac{1}{3}$

equ<sup>n</sup>:  $y - y_1 = m(x - x_1)$

$$y - 3 = -\frac{1}{3}(x - 4)$$

$$3y - 9 = -x + 4 \quad \times 3$$

$$x + 3y - 13 = 0$$

ii)  $d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$   $a=1$   
 $b=3$   
 $c=-13$   
 $x_1=0$   
 $y_1=0$

$$= \frac{|0 + 0 - 13|}{\sqrt{1^2 + 3^2}}$$

$$= \frac{13}{\sqrt{10}} = \frac{13\sqrt{10}}{10} \text{ units}$$

iii) From A to B

translation of: down 2  
across 6

From O - down 2 across 6

$$C = (6, -2)$$

iv)  $d^2 = 6^2 + 2^2$

$$d = \sqrt{40} = 2\sqrt{10} \text{ u.}$$

v)  $A = lh$

$$= 2\sqrt{10} \times \frac{13\sqrt{10}}{10}$$

$$= 26 \text{ u}^2$$

c)  $f(x) = x^3 - 3x^2 - 9x + 22$

i)  $f'(x) = 3x^2 - 6x - 9 = 0$

$$3(x^2 - 2x - 3) = 0$$

$$3(x-3)(x+1) = 0$$

$$x = 3, -1$$

$$f''(x) = 6x - 6$$

$$f''(3) = 12 > 0 \text{ min at } (3, -5)$$

$$f''(-1) = -12 < 0 \text{ max at } (-1, 27)$$

ii) inflexion pt when  $f''(x) = 0$

$$6x - 6 = 0 \Rightarrow x = 1$$

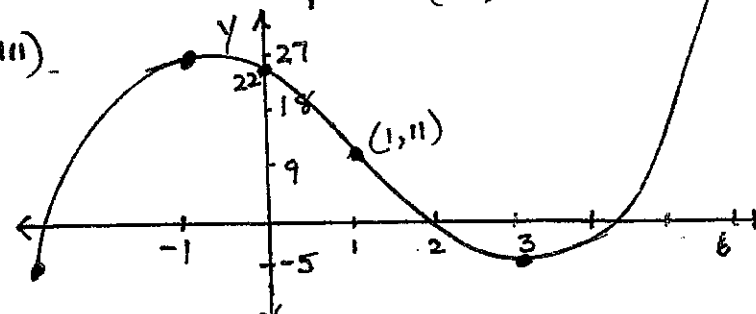
c) ii) cont

test inflexion pt  $\frac{x}{0} \mid \frac{1}{1} \mid \frac{2}{2}$

change in concavity

$$\therefore \text{inflexion pt} = (1, 11)$$

iii)



iv)  $f(6) = 6^3 - 3 \times 6^2 - 9 \times 6 + 22$   
 $= 76$

$$\therefore \text{max value} = 76$$

Question 14

a)  $\lim_{x \rightarrow \infty} \frac{x^4}{5x+5} = \frac{1}{5}$

b)  $\frac{\log_e 22}{\log_e 4} = 2.230$

i)  $\frac{dy}{dx} = \frac{vdu - u dv}{v^2}$   
 $= \frac{x \cdot e^x - e^x \cdot 1}{x^2}$   
 $= \frac{x e^x - e^x}{x^2}$

ii)  $\frac{dy}{dx} = vdu + u dv$   
 $= x \cdot \frac{1}{x} + \log_e 2x \cdot 1$   
 $= 1 + \log_e 2x$

d)  $T_1 = 3^0 = 1$   
 $T_2 = 3^{-1} = \frac{1}{3}$   
 $T_3 = 3^{-2} = \frac{1}{9}$

G.P  
 $a=1$   
 $r=\frac{1}{3}$

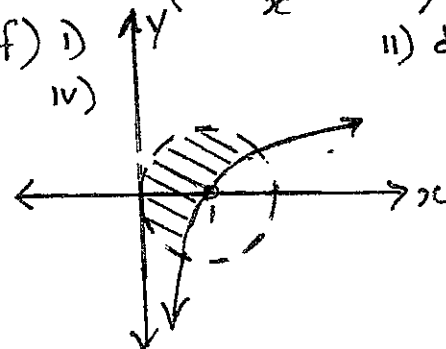
$$S_{\infty} = \frac{a}{1-r} = \frac{1}{1-\frac{1}{3}} = \frac{3}{2}$$

e)  $\frac{1}{2} \log_e (x+3) - \log_e \frac{x}{2}$

$$= \log_e \frac{(x+3)^{1/2}}{x/2}$$

$$= \log_e \left( \frac{2\sqrt{x+3}}{x} \right)$$

f) i) iv)



ii) domain:  
 $x > 0$

Question 14 cont.

f) iii)  $y = \log_e x$

increasing  $\Rightarrow y' > 0$

$$y' = \frac{1}{x} > 0 \text{ when } x > 0 \text{ (domain)}$$

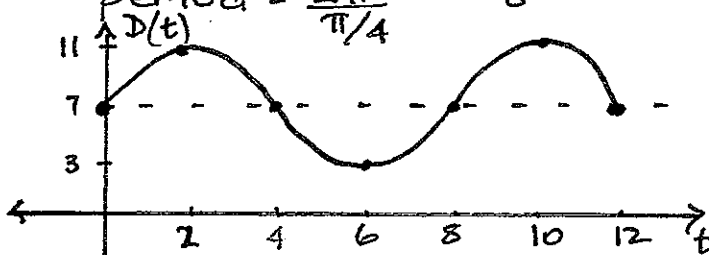
$\therefore y = \log_e x$  is always increasing

Question 15

i)  $D(t) = 4 \sin\left(\frac{\pi}{4}t\right) + 7$

amplitude = 4

period =  $\frac{2\pi}{\pi/4} = 8$



ii) peak at  $t = 2, 10$  years

b) Given ABCD is a rhombus then diagonals bisect its angles  $\therefore \angle BCA = \angle DCA$

ii) FC is common

$\angle BCA = \angle DCA$  from i)

DC = BC (sides of rhombus)

$\therefore \triangle BFC \equiv \triangle DFC$  (SAS)

iii)  $\angle AEB = \angle EBC$

(Alternate  $\angle$ 's AD  $\parallel$  BC)

$\angle AEB = 90^\circ$  (given)

$\angle EBC = 90^\circ = \text{right angle}$

iv) Since  $\triangle BFC \equiv \triangle DFC$

then  $\angle FDC = \angle EBC$

(corresponding  $\angle$ 's of congruent  $\Delta$ 's)

$\angle EBC = 90^\circ$  (from iii)

$\therefore \angle FDC = 90^\circ$

c)  $\int = \left[ \frac{x^2}{2} + kx \right]_0^6 = 30$

$$18 + 6k - 0 = 30$$

$$k = 2$$

15 c) i)  $V = \pi r^2 h \dots \dots (1)$

By Pythagoras

$$r^2 + \left(\frac{h}{2}\right)^2 = R^2 \quad R = 4$$

$$r^2 = 4^2 - \frac{h^2}{4} \dots \dots (2)$$

Sub (2) into (1)

$$V = \pi \left(16 - \frac{h^2}{4}\right) h$$

$$= \pi h \left(\frac{64 - h^2}{4}\right)$$

$$= \frac{1}{4} \pi h (64 - h^2)$$

c) ii)  $V = 16\pi h - \frac{\pi h^3}{4}$

$$\frac{dV}{dh} = 16\pi - \frac{3\pi h^2}{4} = 0 \quad \downarrow \div \pi$$

$$\frac{3h^2}{4} = 16$$

$$h^2 = \frac{64}{3}, \quad h = \pm \frac{8}{\sqrt{3}}$$

Since  $h > 0$ ,  $h = \frac{8\sqrt{3}}{3} \text{ cm}$

Test for max

$$\frac{d^2V}{dh^2} = -\frac{6\pi h}{4}$$

at  $h = \frac{8\sqrt{3}}{3}$ ,  $V'' < 0$

$\therefore$  max Volume at  $h = \frac{8\sqrt{3}}{3} \text{ cm}$

Question 16

a)  $\int = \frac{2-0}{6} [f(0) + 4f(1) + 2f(2) + 4f(3) + f(4)]$   
 $= \frac{1}{3} [\sqrt{2} + 4\sqrt{3} + 2\sqrt{18} + 4\sqrt{83} + \sqrt{258}]$   
 $= 23.11$

b) i)  $S_{n-1} = 3(n-1)^2 - 11(n-1)$   
 $= 3(n^2 - 2n + 1) - 11n + 11$   
 $= \underline{3n^2 - 17n + 14}$

ii)  $T_n = S_n - S_{n-1}$   
 $= 3n^2 - 11n - (3n^2 - 17n + 14)$   
 $= \underline{6n - 14}$

$$16.c) i) y = \log_e(\cos x)$$

$$\therefore y' = \frac{-\sin x}{\cos x} = -\tan x$$

$$ii) \int \tan x \, dx \\ = -\log_e(\cos x) + C$$

$$d) V = \pi \int y^2 \, dx \\ = \pi \int_1^2 \left(\frac{1}{\sqrt{x}}\right)^2 \, dx \\ = \pi \int_1^2 \frac{1}{x} \, dx \\ = \pi \left[ \log_e x \right]_1^2 \\ = \pi [\log_e 2 - \log_e 1] \\ = \pi \log_e 2$$

$$e) A = \int_1^2 e^x - 2 - e^{-x} \, dx \\ = \left[ e^x - 2x + e^{-x} \right]_1^2 \\ = (e^2 - 4 + e^{-2}) - (e - 2 + e^{-1}) \\ = (e^2 - e - e^{-1} + e^{-2} - 2) u^2$$

$$ii) y = e^{-x} \dots (1) \text{ Solve} \\ y = e^x - 2 \dots (2) \text{ simultaneously}$$

Sub (1) into (2)

$$e^x - 2 = e^{-x}$$

$$e^{2x} - 2e^x = 1 \quad \left. \begin{array}{l} \\ \end{array} \right\} \times e^x$$

$$\therefore e^{2x} - 2e^x - 1 = 0$$

$$iii) \text{ let } m = e^x \\ \text{Sub into } e^{2x} - 2e^x - 1 = 0$$

$$\Rightarrow m^2 - 2m - 1 = 0$$

$$a=1, b=-2, c=-1$$

$$= \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$m = \frac{2 \pm \sqrt{(-2)^2 - 4 \cdot 1 \cdot -1}}{2}$$

$$m = \frac{2 \pm \sqrt{8}}{2} = 1 \pm \sqrt{2}$$

$$e^x = 1 + \sqrt{2} \quad \text{or} \quad e^{-x} = 1 - \sqrt{2}$$

$$x = \log_e(1 + \sqrt{2}) \quad \text{or} \quad x = \log_e(1 - \sqrt{2})$$

no solution

$$x = 0.881$$